

Final Exam
MAE C163A/C263A Fall 2021**Name:** _____**UID:** _____

- Make sure you write your name.
- You can use a calculator.
- This exam is open book and notes.
- To receive credit, please be neat and show all key steps of your solutions clearly.
- Please sign the following pledge before beginning the exam:

“I have neither given nor received any unauthorized aid in this test.”

Signature: _____

Problem #	Score
1	/35
2	/30
3	/20
4	/15
Total	/100
Comprehensive	(S/U)

Problem 1 (35 pts)

A 4R manipulator is shown in Figure 1. The nonzero link parameters are $\alpha_1 = -90^\circ$, $d_2 = 1$, $\alpha_2 = 45^\circ$, $d_3 = 1$, and $a_3 = 1$. The mechanism is pictured in the configuration corresponding to $\theta = [0, 0, 90^\circ, 0]^\top$.

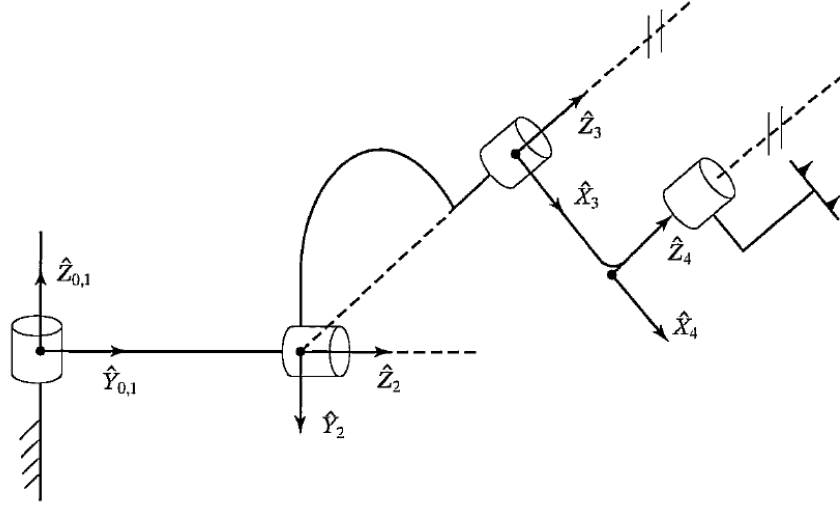


Figure 1: A 4R manipulator shown in the configuration $\theta = [0, 0, 90^\circ, 0]^\top$.

1. Find ${}^0P_{4ORG}$ using forward kinematics. (20 pts)
2. Derive the inverse kinematics for $[\theta_1, \theta_2, \theta_3]$ given ${}^0P_{4ORG} = [p_x, p_y, p_z]^\top$. (Hint: use Pieper's solution or try $p_x^2 + p_y^2 + p_z^2$ for θ_3 ; if you cannot get θ_3 , you can assume it's known and solve for θ_1 and θ_2 for partial credit; 10 pts)
3. Find all values of $[\theta_1, \theta_2, \theta_3]$ such that ${}^0P_{4ORG} = [0, 1, \sqrt{2}]^\top$ and ${}^0P_{4ORG} = [\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}, 0]^\top$. (You are **NOT** required to use the inverse kinematics; 2.5 pts + 2.5 pts)

Problem 2 (30 pts)

Imagine an arm like PUMA 560, except that joint 3 is replaced with a prismatic joint. Assume the prismatic joint slides along the direction of Z_3 shown in Figure 2. However, there is still an offset equivalent to d_3 to be accounted for. The schematic of the robot wrist is shown in Figure 3. The origins of frame 4, 5 and 6 are all at the same point.

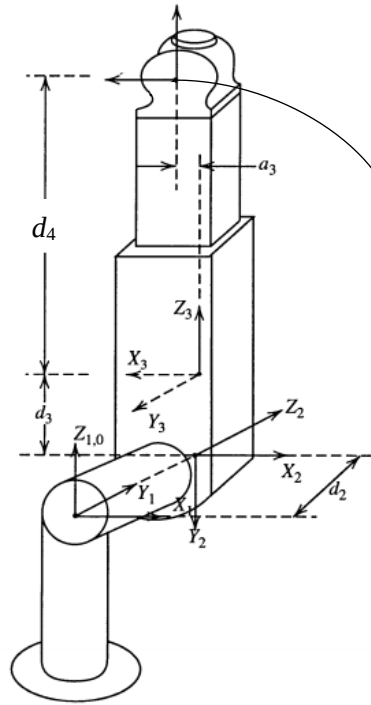


Figure 2: Frame assignments of the manipulator.

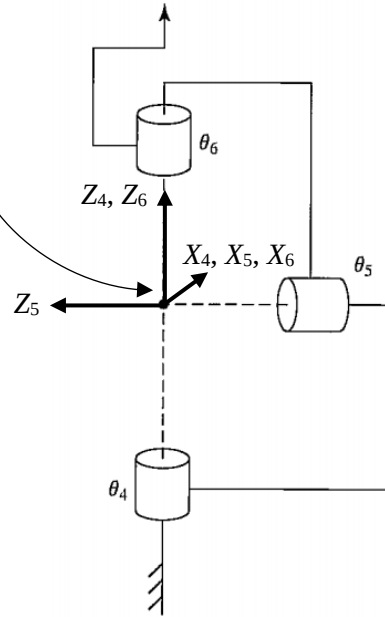


Figure 3: Wrist schematic.

1. Complete the DH table. (10 pts)

i	α_{i-1}	a_{i-1}	d_i	θ_i
1				
2				
3				
4				
5				
6				

2. Write out all the individual transformation matrices between joints and describe how to calculate the overall transformation matrix from frame 0 to frame 6. (You do **NOT** need to calculate 0T_6 ; 10 pts)

3. Given ${}^4T_6 = \begin{bmatrix} r_{11} & r_{12} & r_{13} & 0 \\ r_{21} & r_{22} & r_{23} & 0 \\ r_{31} & r_{32} & r_{33} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, solve for θ_5 and θ_6 symbolically. (10 pts)

Problem 3 (20 pts)

Sketch the **dexterous** workspace of the RPR planar robot. All revolute joints can rotate 360° and self-collision is not considered. Please provide enough dimension information in your sketch.

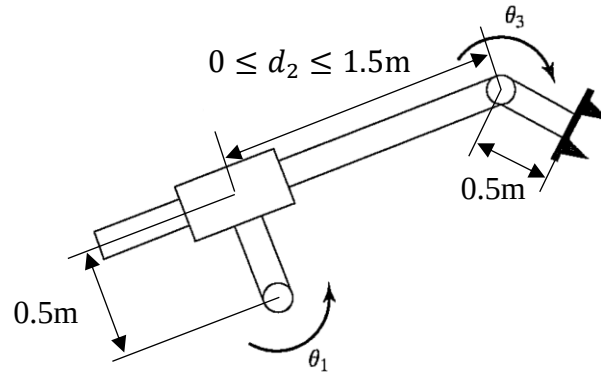
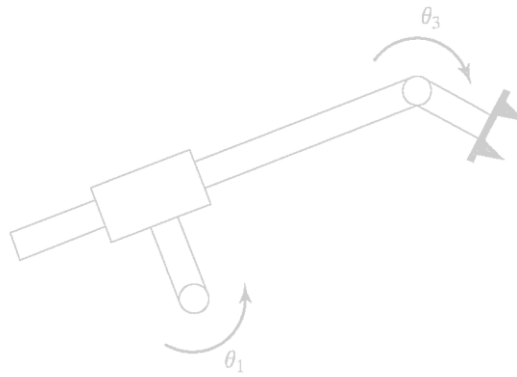


Figure 4: RPR planar robot.



Problem 4 (15 pts)

In Figure 5, two 3R mechanisms are pictured. In both cases, the three axes intersect at a point. The mechanism of (a) has link twists of magnitude 90 degrees. The mechanism of (b) has one twist of ϕ ($< 90^\circ$) in magnitude and the other of $180 - \phi$ in magnitude. The mechanism of (a) can be seen to be in correspondence with Z-Y-Z Euler angles, and therefore we know that it suffices to orient link 3 (with arrow in figure) arbitrarily with respect to the link 0 (base). In other words, the arrow can point to the intersection in any direction. Because ϕ is not equal to 90 degrees, it turns out that the other mechanism cannot orient link 3 arbitrarily.

Sketch the set of orientations that are **unreachable** with the mechanism of (b). Please provide enough dimension information. Note that we assume that all joints can turn 360 degrees (i.e., no limits) and the links may pass through each other if need be (i.e., workspace not limited by self-collision).

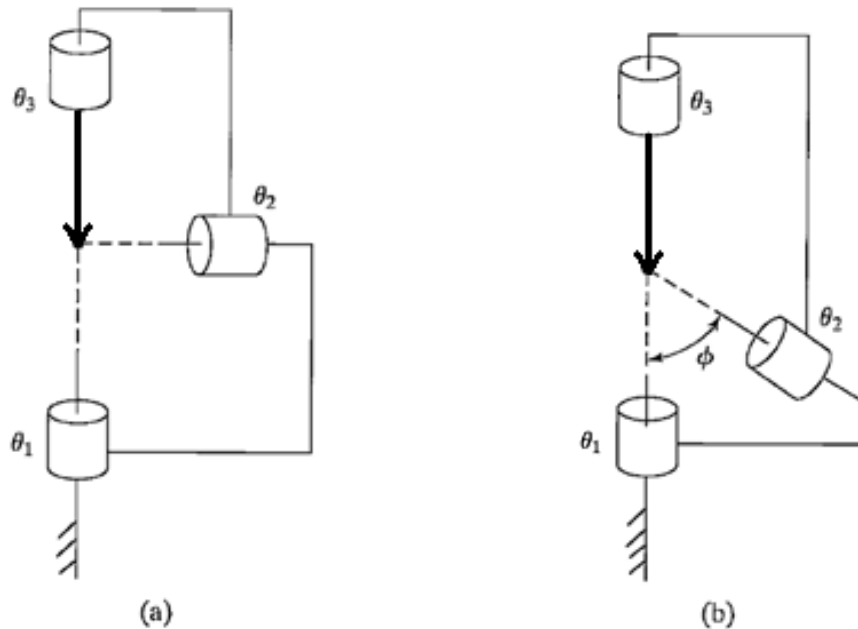


Figure 5: Two 3R mechanisms.

Comprehensive Problem (Optional)

Given ${}^0_3T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$, attach the frames and solve the inverse kinematics for the cylindrical arm below.

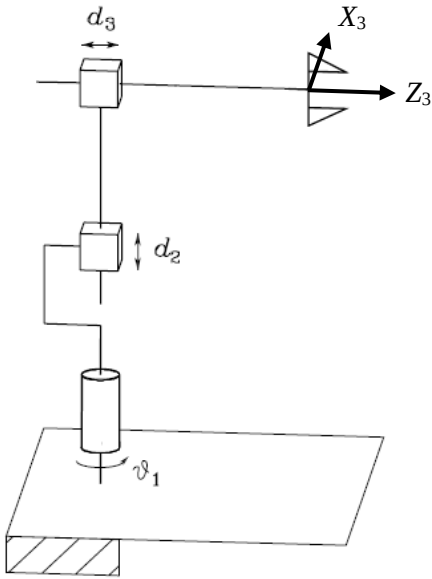


Figure 6: Cylindrical arm.